From Dusty Pages to the Birth of All Things: A Study on the Dual-Track Activation Model of Documentary Heritage Based on Large Language Models

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Documentary heritage is a valuable testament to human history and culture, preserving rich cultural values and historical memories. With growing global interest in cultural heritage, documentary heritage is becoming a key part of the broader cultural heritage system. In 1992, UNESCO launched the Memory of the World (MoW) program to preserve documentary heritage and protect it from oblivion, neglect, decay, and deliberate destruction [1]. While the MoW program has driven global efforts to collect and digitally preserve documentary heritage, it remains in an early stage.

In the age of AI, digitizing cultural heritage helps preserve fragile and endangered documents. However, there is a pressing need to activate the diverse value of documentary heritage to raise public awareness of its importance [2]. Without this activation, these treasures risk becoming mere "dusty pages" locked away in outdated systems.

The development of documentary heritage is complex. Its themes are vast, varied, and often lack clear contextual or causal relationships, creating significant cognitive burdens for administrators, researchers, and the public. This leads to valuable knowledge being buried in overwhelming volumes of resources. Current methods rely mainly on manual analysis and traditional machine learning, but these approaches are insufficient for the specialized, multimodal nature of documentary heritage [3]. Addressing these challenges requires advanced technical capabilities and highly structured, semantically enriched resources.

In 2022, large language models (LLMs), exemplified by ChatGPT, rapidly permeated all areas of human life with their powerful capabilities in cross-task, cross-domain, cross-modal generation, and human-machine interaction [4]. This breakthrough has created both opportunities and challenges for activating documentary heritage. While generative technologies like LLMs have fostered new cultural trends, they also raise higher demands for presenting ancient cultures within documentary heritage. Conversely, the technical advantages of LLMs offer a powerful tool for activating cultural heritage resources. **This study, therefore, focuses on LLMs, proposing the Dual-Track Activation Model of Documentary Heritage,** supported by the "Model Track - Capability Track" framework.

- Model Track

The model track is based on theories of knowledge engineering and cultural computing to construct the " **Extraction - Organization - Service** " activation model for documentary heritage. In the extraction phase, knowledge elements representing the basic features and core connotations of the original resources are extracted, such as time, place, people, events, and objects, along with their semantic relationships. In the organization phase, these knowledge elements are organized based on their semantic relationships and logical structures to form a structured knowledge system. In the service phase, the resulting knowledge system is combined with user needs to build interactive service scenarios. The semantic and logical documentary heritage resources processed through the model track, combined with diverse interactive functions, can not only depict local ele-

ments within the resources but also relate and present a grand historical picture, meeting the diverse needs of the public.

- Capability Track

The capability track establishes a technical pathway based on LLMs for " **Self-Acquisition - Self-Memory - Self-Application** ," allowing LLMs to continuously optimize while supporting the model track. First, without the need for additional training or standards, LLMs traverse and read the raw resources to extract knowledge elements. These fragmented pieces of knowledge are then related based on logical relationships and transformed into a knowledge graph or other semantic network formats. Next, the knowledge graph triples are converted into high-dimensional vector embeddings within the LLM to enable it to "remember" the knowledge contained in the documentary heritage. Finally, the LLM, leveraging its vast knowledge base and newly acquired documentary heritage knowledge, provides diverse knowledge services. Users can interact with the LLM-backed service scenarios to access documentary heritage knowledge. Through continuous interaction and feedback, the LLM undergoes optimization and iteration.

- **Case Study - Suzhou Silk Pattern Knowledge Base Platform**

The Suzhou Silk Archives contains original records from the 19th to the late 20th century documenting the technological research, production management, trade marketing, and awards of Suzhou silk enterprises. The collection consists of 29,592 volumes and is preserved by the Suzhou China Silk Archives. In 2015, this archive was included in the "National Archival Heritage List of China" and was later formally inscribed in the UNESCO Memory of the World Register on October 30, 2017.

This study applies the Dual-Track Activation Model of Documentary Heritage to the Suzhou Silk Archives, using LLM technology to develop a comprehensive silk pattern knowledge base platform. The platform not only provides services for displaying and retrieving silk patterns but also deeply explores the associated knowledge elements, such as relevant people, places, events, and techniques, presenting them in an intuitive form. In doing so, it allows the public to truly perceive the real historical stories, craftsmanship, and silk works that have been sealed in the archives, bringing them back to life.

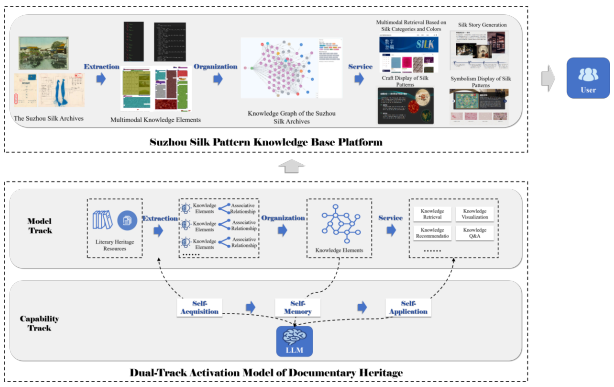

Figure {1} Dual-Track Activation Model of Documentary Heritage

[1] **Dato' Habibah Zon** .(1999): " UNESCO MEMORY OF THE WORLD PROGRAMME.The Asia-Pacific Strategy"<https://web.archive.org/web/20050228192535/http://www.geocities.com/seapavaa/whatsnew/memory.htm>[04.12.2024]

[2] **Abid A** (1998) **:"** Memory of the world-Preserving our documentary heritage", in INSPEL, 32(1): 51-66.

[3] **Feather J** (2006):" Managing the documentary heritage: issues from the present and future", in Preservation management for libraries, archives and museums 1-18.

[4] **Niu Li, Jin Chi, Li Anrunze** (2024): "Application of Large Models in the Digital Transformation of Archival Work: New Opportunities, New Models, and New Changes." (06): 30-38.